A Graphical User Interface for Predicting Laser-Induced Surface Structures and Thermal-Optical Response of Metals

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A Graphical User Interface for Predicting Laser-Induced Surface Structures and Thermal-Optical Response of Metals

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Abstract

Femtosecond pulsed lasers are widely used for precise micro- and nanoscale patterning. Efficient material processing requires accurate knowledge of thermal and optical responses following ultrashort laser irradiation. While many studies have explored laser-induced thermal effects such as ablation, phase transitions and material damage, the combined optical and thermal behavior of thin films with thicknesses near the optical penetration depth remains largely unexplored. Identifying laser conditions that induce laser-induced periodic surface structures (LIPSS) is also crucial for fabricating biomimetic functional patterns for industrial applications. Current analyses often rely on specialized algorithms or expensive software, limiting accessibility. The developed Graphical User Interface provides a user-friendly tool for evaluating LIPSS formation and optothermal responses in various metals (Au, Ag, Cu, Al, Ni, Ti, Cr, stainless steel, W, Pt, Mo) on three substrates (Si, SiO_2, soda-lime glass) including free-standing films. It uses a physical model combining optical absorption and thermal effects via the Two-Temperature Model allowing assessment of photon energy, film thickness, pulse duration, pulse separation and peak fluence. In addition, the interface employs Machine Learning techniques to accelerate the prediction of the metal surface damage threshold, as well as the LIPSS periodicity based on the film thickness and pulse separation.

1. Theoretical model

A. Opto-thermal response

To describe the thermal effects on the material following irradiation with ultrashort laser pulses, a theoretical framework is employed to explore the excitation and thermal response of a double-layered structure (thin metal film on a dielectric material). The simulation algorithm is based on the use of a Two Temperature Model (TTM) that represents the standard approach to evaluate the dynamics of electron excitation and relaxation processes in solids [1]. In this work, for the sake of simplicity, an 1D-TTM is used to describe the thermal effects due to heating of the thin films with laser pulses of wavelength λ_L for a pulse duration τ_p. It is noted that the solution of the TTM in 1D (along the energy propagation direction) is a standard approach used to describe thermal effects after irradiation of solids with femtosecond laser pulses [2, 3]. This is a reasonable choice assuming that, in principle, the laser spot radius is much larger (some tens of micrometers) than the thickness of the irradiated solid; therefore, the laser energy distribution along the lateral direction is considered uniform which means that the heat conduction along that direction is practically infinitesimal. Due to the presence of the substrate, the following set of rate equations is employed [4]

$$\begin{aligned}
C_e^{(m)}\frac{\partial T_e^{(m)}}{\partial t} &= \frac{\partial}{\partial z}\left(k_e^{(m)}\frac{\partial T_e^{(m)}}{\partial z}\right) - G_{eL}^{(m)}\left(T_e^{(m)} - T_L^{(m)}\right) + S^{(m)} \\
C_L^{(m)}\frac{\partial T_L^{(m)}}{\partial t} &= \frac{\partial}{\partial z}\left(k_L^{(m)}\frac{\partial T_L^{(m)}}{\partial z}\right) + G_{eL}^{(m)}\left(T_e^{(m)} - T_L^{(m)}\right) \\
C_e^{(S)}\frac{\partial T_e^{(S)}}{\partial t} &= \frac{\partial}{\partial z}\left(k_e^{(S)}\frac{\partial T_e^{(S)}}{\partial z}\right) - G_{eL}^{(S)}\left(T_e^{(S)} - T_L^{(S)}\right) + S^{(S)} \\
C_L^{(S)}\frac{\partial T_L^{(S)}}{\partial t} &= \frac{\partial}{\partial z}\left(k_L^{(S)}\frac{\partial T_L^{(S)}}{\partial z}\right) + G_{eL}^{(S)}\left(T_e^{(S)} - T_L^{(S)}\right)
\end{aligned} \tag{1}$$

where the subscript 'm' (or 'S') indicates the thin film (or substrate). In Eqs.1, $T_e^{(m)}$ and $T_L^{(m)}$ stand for the electron and lattice temperatures, respectively, of the upper layer. The thermophysical properties of the metal such as the electron $C_e^{(m)}$ (or lattice $C_L^{(m)}$) volumetric heat capacities, electron $k_e^{(m)}\left(=k_{e0}^{(m)}\frac{B_eT_e^{(m)}}{A_e\left(T_e^{(m)}\right)^2+B_eT_L^{(m)}}\right)$ heat conductivity, the electron-phonon coupling strengths $G_{eL}^{(m)}$, A_e, B_e and other model parameters that appear in the first two equations are listed in Table 1 (note A_g=194 Jm^{-3}K^{-1} for Cr). It is emphasised that as heat conduction in metals is, in principle, due to electrons, the thermal conductivity of the lattice is substantially smaller than that of the electron system. To include this large difference, $k_L^{(m)}$ is normally taken either equal to zero ($k_L^{(m)}$=0) [3, 5] or equal to a very small value compared to the electron heat conductivity ($k_L^{(m)}$=0.01$k_e^{(m)}$ is an expression that has been used in other reports [6, 7]). In this work and without loss of generality, we adopt the latter assumption.

The quantity $S^{(m)}$ represents the source term that accounts for the energy that the laser source provides to the metal surface which is sufficient to generate excited carriers on the thin film. As the purpose of the present investigation is to reveal the impact of optically excited *thin* films, it is important to take to into account the following processes: (i) a portion of the energy is absorbed from the material while part of the laser energy is transmitted into the substrate, (ii) the reflectivity and transmissivity of the irradiated material are influenced by a *multiple reflection process* between the two interfaces (air/metal and metal/substrate), (iii) the transmitted energy into the substrate is not sufficiently high to generate excited carriers and therefore, the third equation of Eqs.1 can be removed while the fourth can be simplified by $C_L^{(S)}\frac{\partial T_L^{(S)}}{\partial t}=\frac{\partial}{\partial z}\left(k_L^{(S)}\frac{\partial T_L^{(S)}}{\partial z}\right)+S^{(S)}$ where $T_L^{(S)}$, $C_L^{(S)}$, $k_L^{(S)}$stands for the substrate temperature, volumetric heat capacity and heat conductivity, respectively. The expression for the source term $S^{(m)}$ which is used to excite a metallic surface of thickness d is given from the following formula [5]

$$S^{(m)}=\frac{(1-R-T)\sqrt{4\log(2)}F}{\sqrt{\pi}\tau_p(\alpha^{-1}+L_b)}\exp\left(-4\log(2)\left(\frac{t-3t_p}{t_p}\right)^2\right)\frac{1}{(1-\exp(-d/(\alpha^{-1}+L_b))} \quad (2)$$

where R and T stand for the reflectivity and transmissivity, respectively, L_b corresponds to the ballistic length, α is the absorption coefficient that is wavelength dependent, and F is the peak fluence of the laser beam. The ballistic transport is also included in the expression as it has been demonstrated that it plays significant role in the response of the material [5]. Special attention is required for the ballistic length as in previous works, it has been reported that for bulk materials, L_b in *s/p*-band metals are comparable ($L_b^{(Au)}$=100 nm, $L_b^{(Ag)}$=142 nm, $L_b^{(Cu)}$=70 nm, $L_b^{(Al)}$=46 nm, $L_b^{(Mo)}$=20 nm [5]) while for the *d*-band metals such as Ni, Ti, Cr, stainless steel, Pt, W, it is of the same order as their optical penetration depth [5]. It is noted that the above values of the ballistic length were used for λ_L~ 400-500 nm pulses [5, 8]. The photon energy of the laser beam is expected to influence the ballistic length of the electrons [9] and therefore, the use of the above values for L_b at higher wavelengths (i.e. 1026 nm) could be questionable. Nevertheless, in the absence of appropriate values for L_b at higher photon energies in bibliography, the above values for the ballistic length are used.

The calculation of R and T and the absorbance $A=1-R-T$ are derived through the use of the multiple reflection theory [10]. Thus, the following expressions are employed to calculate the optical properties for a thin film on a substrate (for a *p*-polarised beam)

$$R=|r_{dl}|^2,\quad T=|t_{dl}|^2Re(\tilde{N}_S),\quad r_{dl}=\frac{r_{am}+r_{mS}e^{2\beta j}}{1+r_{am}+r_{mS}e^{2\beta j}},\quad t_{dl}=\frac{t_{am}t_{mS}e^{\beta j}}{1+r_{am}+r_{mS}e^{2\beta j}},\quad \beta=2\pi d/\lambda_L \quad (3)$$

$$r_{CD}=\frac{\tilde{N}_D-\tilde{N}_C}{\tilde{N}_D+\tilde{N}_C},\quad t_{CD}=\frac{2\tilde{N}_C}{\tilde{N}_D+\tilde{N}_C} \quad (4)$$

where the indices $C=a,m$ and $D=m,S$ characterise each material ('a', 'm', 'S' stand for 'air', 'metal', 'substrate', respectively). The complex refractive indices of the materials such as air, metal and substrate are denoted with $\tilde{N}_a=1$, $\tilde{N}_m=Re(\tilde{N}_m)+Im(\tilde{N}_m)j$, $\tilde{N}_S=Re(\tilde{N}_m)+Im(\tilde{N}_m)j$. It is noted that a Drude-Lorentz model is used to obtain the dielectric function for each metal based on the analysis by Rakic *et* al. (where both interband and intraband transitions are assumed) [11]. As the optical parameters of an excited material do not remain constant during the excitation process [12], to introduce the transient change, a temporally varying expression of the dielectric function is provided by including a temperature dependence on the reciprocal of the electron relaxation time τ_e (i.e. $\tau_e=\left[A_e\left(T_e^{(m)}\right)^2+B_eT_L^{(m)}\right]^{-1}$) [13]. The values of the refractive indices of the metals in this study (at 300 K are given in Table 1).

The set of equations Eqs.1-4 is solved by using an iterative Crank-Nicolson scheme based on a finite-difference method. It is assumed that the system is in thermal equilibrium at t=0 and, therefore, $T_e^{(m)}(z,t=0)$=$T_L^{(m)}(z,t=0)=300$ K. A thick substrate is considered (i.e. $k_L^{(S)}\frac{\partial T_L^{(S)}}{\partial z}=0$) while adiabatic conditions are applied on the surface of the metallic surface (i.e. $k_e^{(m)}\frac{\partial T_e^{(m)}}{\partial z}=0$). Finally, the

following boundary conditions are considered on the interface between the top layer and the substrate: $k_L^{(m)} \frac{\partial T_L^{(m)}}{\partial z} = k_L^{(S)} \frac{\partial T_L^{(S)}}{\partial z}, k_e^{(m)} \frac{\partial T_e^{(m)}}{\partial z} = 0, T_L^{(m)} = T_L^{(S)}$.

	Metal										
Parameter	**Au**	**Ag**	**Cu**	**Al**	**Ni**	**Ti**	**Cr**	**Steel (100Cr6)**	**W**	**Mo**	**Pt**
$\widetilde{N}_m$	DL [11]	DL [11]	DL [11]	DL [11]	DL [11]	DL [11]	DL [11]	DL [13]	DL [11]	- [11]	DL [11]
$G_{eL}^{(m)}$ [$Wm^{-3}K^{-1}$]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	42×10^{16} [5]	Ab-Initio [15]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]
$C_e^{(m)}$ [$Jm^{-3}K^{-1}$]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]	$A_g T_e^{(m)}$ [5]	Ab-Initio [15]	Ab-Initio [14]	Ab-Initio [14]	Ab-Initio [14]
$C_L^{(m)}$ [$\times10^6$ $Jm^{-3}K^{-1}$]	2.48 [5]	2.5 [7]	3.3 [5]	2.4 [16]	4.3 [5]	2.35 [17]	3.3 [5]	3.27 [15]	2.5 [17]	2.33 [5]	2.8 [15]
$k_{e0}^{(m)}$ [$Wm^{-1}K^{-1}$]	318 [5]	428 [7]	401 [5]	235 [16]	90 [5]	21.9 [17]	93.9 [5]	46.6 [15]	21.9 [17]	138 [5]	72 [15]
A_e [$\times10^7$ $s^{-1}K^{-2}$]	1.18 [7]	0.932 [7]	1.28 [7]	0.376 [16]	0.59 [7]	1 [17]	7.9 [6]	0.98 [15]	1 [17]	1 [6]	1 [15]
B_e [$\times10^{11}$ $s^{-1}K^{-1}$]	1.25 [7]	1.02 [7]	1.23 [7]	3.9 [16]	1.4 [7]	1.5 [17]	13.4 [6]	2.8 [15]	1.5 [17]	1.5 [6]	1.5 [15]
T_{melt} [K]	1337 [18]	1234 [18]	1357 [18]	933 [18]	1728 [18]	1941 [18]	2180 [18]	1811 [18]	1941 [18]	2896 [18]	2045 [18]

Table 1: Optical and thermophysical properties of materials (*DL* stands for Drude-Lorentz model).

B. Onset of formation of Laser Induced Periodic Surface Structures

One important aspect in material processing is the fabrication of subwavelength structures via the use of laser pulses. Although there is a plethora of different types of structures that are normally generated via the usage of lasers [19-21], the emphasis in this work will be centred on the formation of periodic surface structures on metals. As excitation of surface plasmons (SP) and their interference with the incident beam is regarded as the predominant mechanism of periodic structure formation in metals [22, 23], the periodicities of the (allowed) excited electromagnetic surface modes could provide an estimate of the expected sub-wavelength structures on the irradiated material. Excitation of surface modes, typically associated with surface plasmons in metals, requires a non-flat surface. To represent this condition, an initial surface roughness is assumed, arising from the formation of bumps or cavities caused by mass displacement following a phase transition. Although a rigorous analysis would require computationally intensive multiscale, multiphysics modelling including electromagnetic and fluid dynamics simulations in addition to the evaluation of excitation and thermal effects, an approximate criterion can be applied. More specifically, the relevant conditions are estimated by assuming a laser fluence of approximately ~1.1**DT* where *DT* corresponds to the damage threshold, defined as the minimum fluence required to induce melting. The selected fluence is aimed to generate a sufficiently large modified volume that results in surface irregularities (i.e., roughness) which are necessary for surface plasmon excitation. It is noted that while the SP periodicity (and, eventually, the LIPSS periodicity oriented perpendicularly to the laser polarisation) is dependent on the number of pulses that irradiate the material [19, 22], to avoid a time consuming analysis, we focus on the onset of SP generation conditions. Thus, it is assumed that *only one* pulse following the presence of a rough pattern is necessary to generate a LIPSS-covered topography. On the other hand, it is emphasized that the SP wavelengths are calculated via the following schemes:

a. For bulk Materials (thickness >~70 nm)

The SP wavelength Λ is derived via the use of the expression

$$\Lambda = \frac{\lambda_L}{Re\sqrt{\frac{\varepsilon_a \varepsilon_m}{\varepsilon_a + \varepsilon_m}}} \tag{5}$$

where $\varepsilon_a, \varepsilon_m$ correspond to the dielectric parameter at $\lambda_L$of air (=1) and the metal, respectively. (**Important:** It is assumed that the thickness does *not* vary the dielectric parameter.)

b. For thin films (thickness comparable to the optical penetration depth)

The SP wavelength Λ is derived via the use of the dispersion relation

$$exp(-2k_m d) = \frac{k_m/\varepsilon_m + k_a/\varepsilon_a}{k_m/\varepsilon_m - k_a/\varepsilon_a} \times \frac{k_m/\varepsilon_m + k_S/\varepsilon_S}{k_m/\varepsilon_m - k_S/\varepsilon_S} \quad (6)$$

where d is the thickness of the film, $k_j = \sqrt{\tilde{\beta}^2 - \varepsilon_j k_0^2}$ (j= 'a' 'm', 'S', for 'air', 'metal', 'substrate'), ε_j stands for the dielectric permittivity of the j material, k_0 (=2π/λ_L) is the free-space wavenumber at the laser wavelength and $\tilde{\beta}$ is the propagation constant of the SP. Thus, the above equation yields the supported SP solutions; each solution is associated with an SP wavelength provided by $\Lambda = \frac{2\pi}{Re(\tilde{\beta})}$.

2. Simulations

I. Calculations based on the Physics-Based Model

The algorithm's execution considers both the thermophysical and optical parameters (i.e. dielectric parameter) of the irradiated metal/substrate complex. The selection of the materials was based on considering the most common metals and substrates while it is aimed to be extended soon to other metals and configurations (multi-layered materials, etc). Similarly, an upcoming code update will enable users to study the dynamics of wide-bandgap materials, including semiconductors, oxides, dielectrics, and polymers.

A minimal intervention is required from the user, mostly, to introduce the refractive index and extinction coefficient for some materials. Upon introducing the required properties of the materials and the laser conditions, the algorithm calculates, firstly, the damage threshold *DT*, before deriving a series of figures that include the thermal effects (transient electron and lattice temperature), reflectivity, absorptivity and transmissivity and an evaluation of the SP wavelength for a fluence ~1.1**DT*.

Steps to perform the evaluation of the thermal phenomena, optical parameters and periodicity of the LIPSS via the interface

In order to use the computational tool, the user has to select, firstly,

a) the laser conditions (i.e. laser wavelength, pulse duration, pulse delay (in case of double pulses))
b) metal (type of metal and thickness)
c) substrate.

a) the 'laser wavelength' must be edited in 'nm'.
The 'pulse duration' (Full Width at Half Maximum of a Gaussian-like pulse) must be edited in 'fs'.
The 'pulse delay' (time separation between subsequent pulses) must be edited in 'fs'.
The value in 'Specify the end of the time interval (starts at 0):' (time by which evolution of the optical and thermal effects are evaluated) must be edited in 'ps'.
The 'thickness of material' (thickness of the metal) must be edited in 'nm'.

b) Calculations are currently performed for the class 'metal': 'Au', 'Ag', 'Cu', 'Al', 'Ni', 'Cr', 'Ti', 'Steel' (100Cr6), 'W', 'Mo', 'Pt'.

c) Four types of materials are considered as 'Substrate': 'Silicon', 'Fused Silica', 'Soda Lime Silica Glass', 'Air'.

i. Input/Output

ii. Range of the input variables

- For the 'pulse duration' (τ_p), faster solutions are obtained for values between 5 fs and up to some hundreds of ps (or few ns). **Important:** In general, τ_p should be larger than 100 fs. Although shorter pulse durations can still be evaluated via the interface, results might not be physical as the employment of TTM to describe ultrafast dynamics at shorter pulse durations is generally *not* valid. Execution of the code is more time consuming for long pulses. In principle, the code was derived especially, to illustrate thermal and optical effects for ultrashort pulses. Although the code, in effect can be used for 'ns' long pulses, despite the long execution time, a significantly faster code is planned to be incorporated soon to address phenomena at the ns-scale.
- For the 'pulse delay' (t_{delay}), faster solutions are obtained for values between 100 fs (where the TTM is valid) and up to some hundreds of ps. Pulse delays in the ns scale could also be tried. It might take longer.
- For the 'Specify the end of the time interval (starts at 0):' This parameter defines the final time of the interval over which the evolution of the various physical quantities is calculated and displayed. As a general guideline, the following expression can be used to estimate a suitable value that clearly captures the system dynamics: $(2*6*\tau_p + t_{delay} +80000)/1000$. If the time delay is zero, the simplified expression $6*\tau_p+40000)/1000$ is used instead. It should be emphasized that the 'maximum time' value is not restrictive. The suggested expressions provide typical values that adequately represent the relaxation time range, but shorter or longer intervals may also be selected depending on the specific case. For example, in double-pulse irradiation with 170 fs pulse duration and a pulse separation of 2 ps, a representative value for 'Specify the end of the time interval (starts at 0):' could be approximately 100 ps. Although longer time intervals can be chosen, they may increase the computation time. In such cases, the early-stage dynamics may not be clearly visible in the plotted figures; however, detailed inspection of the output data files allows closer examination of any user-defined time region.
- For the 'thickness of material', values between ~10-15 nm (marginally lower than the optical penetration depth) and <~1 μm (in principle, some hundreds of nm where the metal starts to behave as a bulk solid).
- For the 'laser wavelength', the range of the wavelength values of the laser beam that irradiates the metal is taken on data from paper of Rakic et al [11] (see also range in RefractiveIndex.INFO - Refractive index database considering the choice 'Rakic…'. In general, the wavelength range that has been tested so far for the purposes of this interface lie between 248 nm and up to some micrometers.

IMPORTANT:

- ✓ for 'Steel', for one wavelength ('1026 nm'), the optical properties of the material have already been incorporated into the model and no intervention from the user is required. For other laser wavelengths, the user has to enter: 'the real part of the refractive index' and the 'imaginary part of the refractive index (extinction coefficient)'.
- ✓ Similarly, for 'Mo', for one wavelength ('1026 nm'), the optical properties of the material have already been incorporated into the model and no intervention from the user is required. For other laser wavelengths, the user has to enter: 'the real part of the refractive index' and the 'imaginary part of the refractive index (extinction coefficient)'. Data should be collected from either the above database or from any other source.
- ✓ For 'Silicon' substrate, for four wavelengths ('248 nm', '515 nm', '800 nm', '1026 nm') the refractive index of the substrate has already been incorporated and no intervention from the user is required. For other laser wavelengths, the user has to enter: 'the real part of the refractive index' and the 'imaginary part of the refractive index (extinction coefficient)'. Data should be collected from either the above database or from any other source.
- ✓ Other substrates that are used in this interface are 'SiO_2', 'Soda Lime glass', 'Air' (i.e. free standing film). The refractive indices for those substrates are taken from the above database automatically.

iii. Execution of the Code

Following entering the above data, the user has to press the button 'Calculate' in order to start the evaluation of the evolution of the temperatures of the electrons, lattice and optical parameters inside the irradiated material as well as the impact of the substrate. **Results are sent to the user via Email.**

iv. Output

The execution of the code leads firstly to the creation of a directory named after the metal that has been selected (for example: 'Cr'). Inside the directory, a series of figures (in .png format) are produced. To facilitate data management and enable further analysis, the text files (.txt format) used to generate the figures are also provided. Finally, a file (in .txt format) that summarises the conditions is also produced:

a. Text Files

- 'Conditions_for_results.txt'**:** The text file that is produced provides a summary of the laser conditions (i.e. fluence required to melt the material called 'Damage threshold') and material parameters; the calculated temperature for the evaluated damage threshold is compared with the melting temperature; Results shown in the file are listed in the following format:

  Material: Cr
  Substrate: SiO2
  Pulse duration (s) = 1.700000e-13
  Thickness of First Layer (m) = 1.000000e-07
  Pulse delay between pulses(s) = 0
  Wavelength (m) = 1.030000e-06
  Damage Threshold (J/cm2) = 8.691264e-02
  Maximum Temperature on surface (K) = 2.179188e+03
  Melting Temperature (T_melt) (K) = 2180

- 'Conditions_materials_results.txt'**:** The text file that is produced provides a summary of the laser conditions (assuming fluence ~1.1*DT*) and material parameters; the maximum lattice temperature on the surface for that fluence is calculated; the thickness of the molten volume is calculated; the ablation temperature is set to be equal to $0.95*T_{boiling}$ or $0.95*T_{critical\ temperature}$; Results shown in the file are listed in the following format:

  Material: Cr
  Substrate: SiO2
  Thickness of First Layer (m) = 1.000000e-07
  Thickness of Substrate (m) = 1.000000e-07
  Pulse delay (s) = 0
  Wavelength (m) = 1.030000e-06
  Fluence (J/cm2) = 9.212739e-02
  Maximum Temperature on surface (K) = 2.308049e+03
  Melting Temperature (T_melt) (K) = 2180
  Thickness of Molten material (i.e. if Lattice Temperature in the layer is larger than Tmelt) (m) = 9.000000e-09
  Ablation Temperature (T_ablation) (K) = 2.796800e+03

  Considering the above conditions (in 'Conditions_materials_results.txt') the following .txt files are produced:

- 'Stack depth_(m).txt': A column that indicates the stack thickness in discretized form.
- 'Metal depth_(m).txt': A column that indicates the metal thickness in discretized form.
- 'Time_Intensity_reflectivity.txt': A three column matrix providing the 'Time', 'Intensity' (Arbitrary Units), 'Reflectivity'.
- 'Time_Intensity_tranmissivity.txt': A three column matrix providing the 'Time', 'Intensity' (Arbitrary Units), 'Transmissivity'.
- 'Time_Intensity_absorptivity.txt': A three column matrix providing the 'Time', 'Intensity' (Arbitrary Units), 'Absorptivity'.
- 'Time_Intensity.txt': A two column matrix providing the 'Time', 'Intensity' (Arbitrary Units).
- 'Te_upper_layer_(t,z)_(K).txt': A matrix that provides the T_e field inside the metal (rows and columns correspond to the depth and time, respectively).
- 'TL_stack_(t,z)_(K).txt': A matrix that provides the T_L field inside the stack (rows and columns correspond to the depth and time, respectively).

- ‘TL_upper_layer_(t,z)_(K).txt’: A matrix that provides the T_L field inside the metal (rows and columns correspond to the depth and time, respectively).
- ‘Time_Te_TL.txt’: A three column matrix giving the ‘Time’, ‘T_e’, ‘T_L’.

b. Figure files

- ‘Reflectivity.png’: The evolution of the reflectivity of the stack is illustrated in Fig.1a (*red line*). The *black dashed line* shows the evolution of the laser intensity (in arbitrary units).

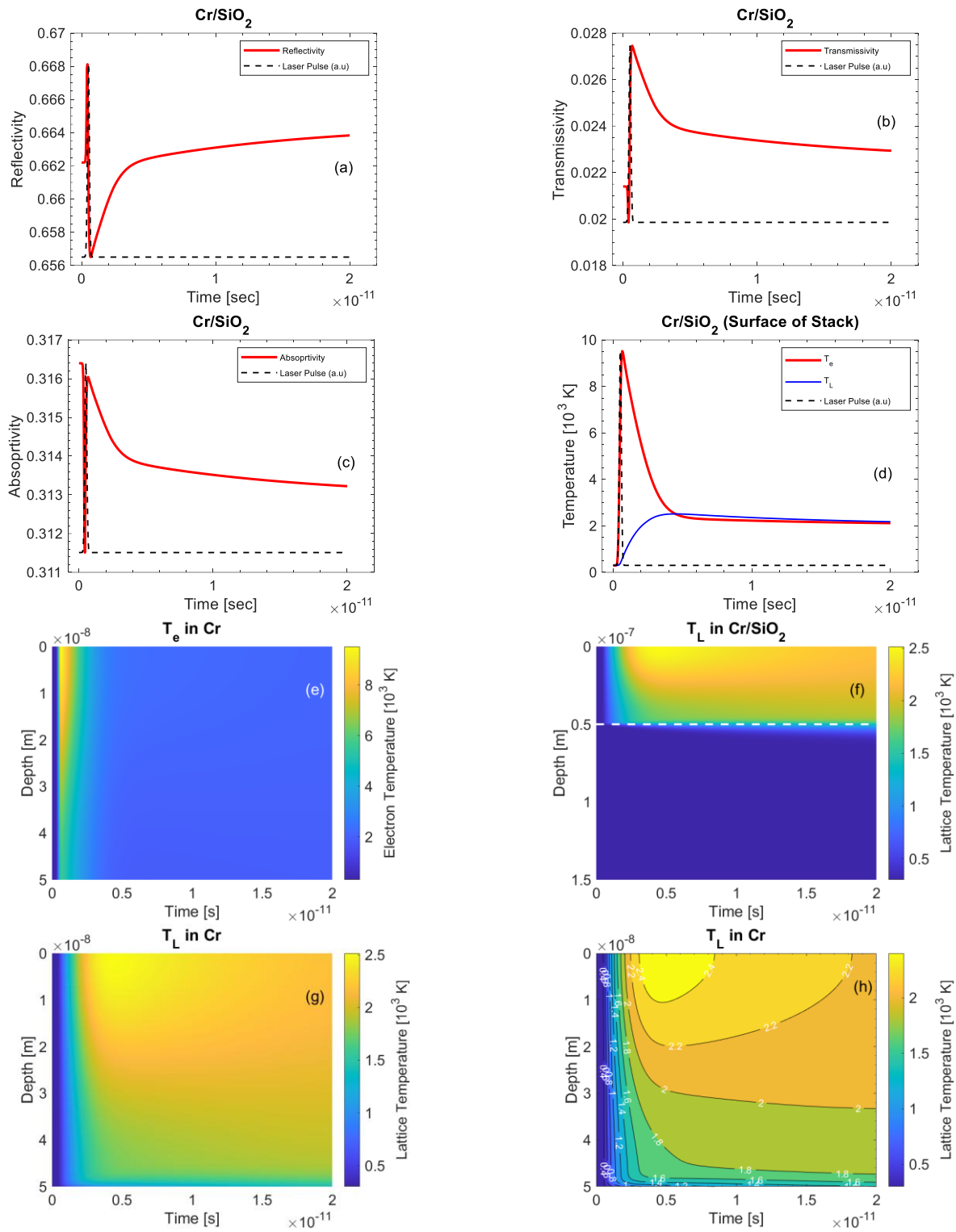

Figure 1: Evolution of (a) Reflectivity, (b) Transmissivity, (c) Absorptivity. (d) Electron Temperature and lattice temperature evolution, (e) Electron temperature evolution inside the depth of Cr. Lattice temperature evolution inside the stack (f), metal (g). (h) Contour plot of the Lattice temperature evolution inside the metal (lines show the isotherms).

- 'Transmissivity.png': The evolution of the transmissivity of the stack is illustrated in Fig.1b (*red line*). The *black dashed line* shows the evolution of the laser intensity (in arbitrary units).
- 'Absorptivity.png': The evolution of the absorptivity of the stack is illustrated in Fig.1c (*red line*). The *black dashed line* shows the evolution of the laser intensity (in arbitrary units).
- 'Te_Tl_t.png': The evolution of the electron temperature T_e (*red line*) and the lattice temperature T_L (*blue line*) on the surface of the metal is illustrated in Fig.1d. The *black dashed line* shows the evolution of the laser intensity (in arbitrary units).
- 'Te_t_z.png': The evolution of the electron temperature T_e inside the metal's volume is illustrated in Fig.1e.
- 'TL_t_z_stack.png': The evolution of the lattice temperature T_L inside the stack's (metal/substrate) volume is illustrated in Fig.1f. The *white dashed line* defines the interface between the metal and the substrate.
- 'TL_t_z_upper.png': The evolution of the lattice temperature T_L inside the metal's volume is illustrated in Fig.1g.
- 'TL_t_z_upper_layer_contour.png': The evolution of the lattice temperature T_L inside the metal's volume is illustrated in Fig.1h. Contours are included in the figure that shows the isotherms.

Finally, in regard to the calculation of the induced LIPSS periodicity, relevant information is provided in the following text

- 'Conditions_for_LIPSS_results.txt'**:** The text file that is produced provides a summary of the optical properties of the excited film (real and imaginary part of the dielectric parameter), the refractive index of the substrate and the SP wavelength. Results shown in the file are listed in the following format:

  Maximum Re(epsilon) of metal = 14.5980
  Maximum Im(epsilon) of metal = 40.4697
  Refractive index of substrate = 1.4500
  LIPSS Periodicity (nm) = 1033.7172

Further assistance

For more information on the theoretical model read Refs.[24-27] or contact George D. Tsibidis (Institute of Electronic Structure and Laser) at tsibidis@iesl.forth.gr. Furthermore, it is noted that a complementary module of the code has been developed to evaluate the damage thresholds (i.e. minimum laser peak fluence that leads to material melting). This module is on the NFFA Tool.

II. Machine Learning assisted calculation of Damage Threshold and LIPSS periodicity

A feed-forward neural network model has been developed to predict the damage threshold for each material in the study on the SiO_2 substrate, given the film material and thickness, and the pulse separation, using simulated data. The simulated data were produced with a constant laser wavelength of 1026 nm and a pulse duration of 170 ps. The model takes as input the film material, its thickness, and the pulse separation, and predicts the damage threshold, i.e., the fluence at which the lattice temperature reaches the substrate's material melting temperature. The material information enters the model as conditioning inputs through its characteristic thermophysical properties, ρ_l (liquid-phase density), ρ_s (solid-phase density), C_{Ll}, C_{Ls}, A_e, B_L, k_{e0}, T_{cr}, *and* T_{melt}. The predicted damage threshold is then used to determine the fluence (see above) which allows the generation of a sufficiently thick molten layer that can lead to LIPSS formation. In this case, machine learning is used to accelerate the calculation of the damage threshold by bypassing the time-consuming numerical solution of the differential equations guiding the system.

i. Execution of the Code

Following entering the above data, the user has to press the button 'Run Prediction' in order to start the evaluation of the evolution of the temperatures of the electrons, lattice and optical parameters inside the irradiated material as well as the impact of the substrate. **Results are sent to the user via Email.** Results include information about the laser conditions, material, damage threshold, optical parameters of the material and LIPSS periodicity.

Further assistance

For questions regarding the Machine Learning approach, contact Georgios D. Barmparis (Institute of Electronic Structure and Laser) at barmparis@iesl.forth.gr

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